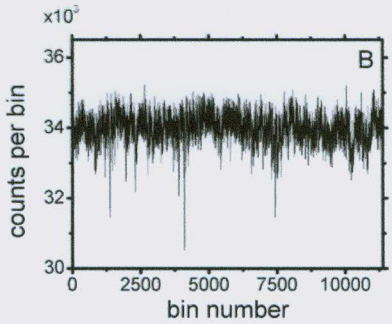
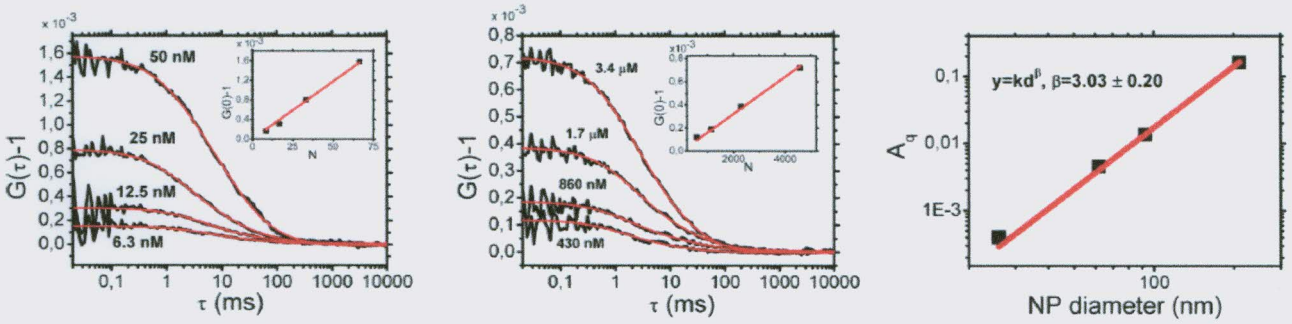
Physical Chemistry Cume – January 25, 2013

Stefan Wennmalm and Jerker Widengren. Interferometry and Fluorescence Detection for Simultaneous Analysis of Labeled and Unlabeled Nanoparticles in Solution. *J. Am. Chem. Soc.*, **2012**, *134,* 19516–19519

Figure 1B from the paper is given below:

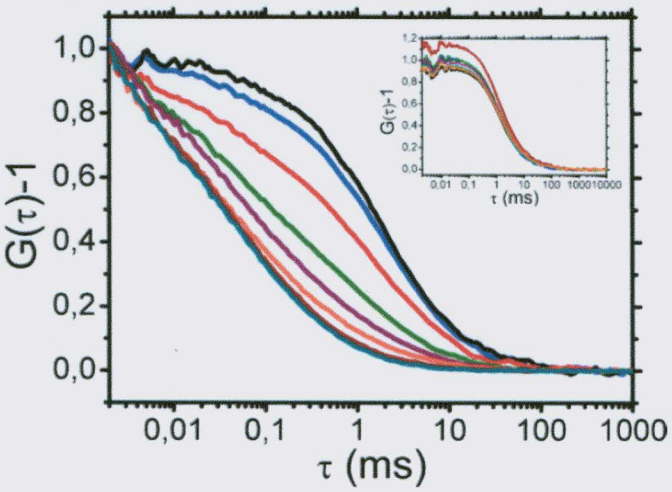


1) Explain how the authors used data such as figure 1B to obtain figure 2 (below). In doing so, explain the significance/meaning of A_q.
(30 points)



2) How did the authors conclude that the negative fluctuations in the intensity of figure 1B were caused by scattering interference rather than by pure scattering? What is the source of this interference?
(20 points)

3) The smallest particles that the authors measured were about 24 nm. What are the factors that determine the minimum size of particle that can be measured using this interferometry technique (scattering interference correlation spectroscopy, SICS), and how can smaller particles (e.g. biomolecules of 2-3 nm) be measured.
(20 points)

(Question 4 on next page)

4) Explain how SICS is combined with Fluorescence Correlation Spectroscopy (FCS) to measure the binding constant, K_d, of a (fluorescent) ligand to the nanoparticles (figure 4 from paper, below).
(30 points)